Reg. No. | | | | | | | | | | |

[22MSCH3H02]

THIRD SEMESTER UNIVERSITY EXAMINATION, NOVEMBER 2023

APPLIED CHEMISTRY-I

Time: 03 HOURS

Maximum Marks: 50

SECTION – A

1. **Answer any FOUR of the following questions.** 4×5=20

   a. Write a note on the preparation of graphene and its characterization techniques.

   b. Discuss the synthesis and applications of polycarbonates.

   c. What are pigments and explain what are the properties to be taken care of while selecting pigments.

   d. What is coagulation? Give a detailed note on the coagulation process.

   e. i. Write any two advantages and limitations of homogeneous catalysis.

      ii. How to prevent the poisoning of the catalyst?

SECTION – B

2. **Answer any TWO of the following questions.** 2×9=18

   a. Explain the difference between X-ray Absorption Spectroscopy and X-Ray Fluorescence spectroscopy.

   b. Write a detailed note on Additives and surface-active agents for paints- driers, corrosion inhibitors, plasticizers, defoamers, UV stabilizers, adhesion promoters, anti-skinning agents and texturizers.

   c. Discuss the theory of homogeneous catalysis and write any two advantages and limitations.

SECTION – C

3. **Answer the following question.** 12

   a. i. What are the different types of carbon allotropes? Define three types with examples.

      ii. What are polyamides? Write a note on nylons.

      iii. Write a note on Donnan membrane equilibrium with illustrations.